

Prequel

Platform Reference Document — Canada Edition

Hiring-signal intelligence, built from Canadian public regulatory data

Working draft · v0.2 · September 2026

This document is a living reference for the Prequel platform: the problem it addresses, the product vision across all phases, the data architecture, technical approach, and known risks. It is intended to be updated as the platform evolves beyond v1.

1. The Problem

Job search tools today are almost entirely reactive: they help people search, filter, or tailor applications against postings that already exist. But a job posting is a lagging indicator — by the time a role is public, hiring intent, budget approval, and often an internal candidate shortlist already exist. Job seekers are structurally last to know.

Meanwhile, a wide range of hiring-intent signals are generated well before a posting goes live, and almost none of them are surfaced to Canadian job seekers in usable form:

- Labour Market Impact Assessment (LMIA) filings, published by ESDC, that reveal exactly which employers, occupations (NOC codes), and business locations are staffing via the Temporary Foreign Worker Program.
- Statistics Canada's Job Vacancy and Wage Survey (JVWS), a quarterly macro signal on vacancy counts and wages by industry and region — useful for validating trends seen in employer-level data.
- Public company disclosures on SEDAR+ (Canada's system for public company filings) that reference headcount, restructuring, or segment investment.
- Provincial mass-termination notices (e.g. under Ontario's Employment Standards Act) that reveal contraction — a patchwork signal, but useful where available.
- Engineering and product activity (commit velocity, new repos, conference presence) that correlates with team growth before a requisition opens.

Prequel's premise: if hiring activity leaves a public trail before it becomes a posting, a tool built on that trail gives Canadian job seekers a structural advantage that no amount of resume tailoring or interview prep can replicate.

2. Product Vision

Prequel is a hiring-signal intelligence platform. It ingests public regulatory and market data, normalizes it into company- and role-level hiring trends, and surfaces it to job seekers as an early-warning layer that sits upstream of traditional job boards.

The full platform is built in phases. Each phase adds a data source and a corresponding layer of insight:

Phase	Data Source	Core Insight	Status
v1	ESDC Positive LMIA employer list	Which employers/occupations (NOC)/locations are actively staffing via the TFWP, and at what scale	Launched
v2	StatCan Job Vacancy & Wage Survey	Macro-level hiring and wage trends by industry and region	Launched
v3	SEDAR+ filings & earnings	Segment-level investment and restructuring language tied to headcount, for public companies	Launched
v4	GitHub / engineering activity	Team growth signal for tech roles specifically, ahead of employer	Expanding
v5	Org-structure reconstruction	Who actually holds hiring power on a target team, not just who reports to whom	Expanding

Each phase is additive and independently useful — the platform is designed so v1 ships as a working, valuable product on its own, not a fragment waiting for later phases.

3. Target Users

- **Primary (v1):** Job seekers, including international students and new grads, in occupations where the Temporary Foreign Worker Program (LMIA) dataset provides dense coverage — notably tech, skilled trades, and healthcare.
- **Secondary (v2+):** Job seekers in any field once StatCan JVWS, provincial layoff notices, and SEDAR+ signals broaden coverage beyond TFWP-heavy occupations.
- **Future:** Recruiters and career coaches who want early market intelligence on which Canadian employers/teams are expanding.

Known limitation: the LMIA dataset only covers roles employers sought to fill via the Temporary Foreign Worker Program — it excludes employer records that include personal names (e.g. many caregiver postings), and it does not capture domestic-only hiring at employers who never file an LMIA. This is explicitly acknowledged rather than hidden — v1's value proposition is honest about who it serves best.

4. Data Architecture

4.1 Ingestion

- **ESDC Positive LMIA Employers List:** quarterly bulk CSV/XLSX downloads via open.canada.ca, broken out by program stream, NOC (National Occupational Classification) code, and business location. No auth, no scraping required — direct file ingestion.
- **StatCan Job Vacancy and Wage Survey:** quarterly data available via Statistics Canada's data tables/API, providing vacancy counts and wages by industry (NAICS) and region — a macro cross-check against employer-level LMIA trends.
- **Provincial mass-termination notices:** patchwork by province (e.g. Ontario's Employment Standards Act requires notice to the Ministry of Labour for mass terminations); no unified federal system. Requires a per-province review of what's publicly accessible before building a scraper/normalizer layer in v2.
- **SEDAR+ filings:** Canada's system of record for public company disclosures; used for full-text search of headcount/restructuring language tied to business segments.
- **GitHub activity:** public REST API, rate-limited; used only as a secondary confidence signal, not a primary source.

4.2 Normalization

The hardest and most valuable engineering problem in this platform is entity resolution: collapsing inconsistent employer name strings ("Shopify Inc.", "Shopify Inc", "SHOPIFY INC.") into a single canonical company entity, and mapping NOC occupation codes into consistent, job-seeker-friendly role

families (NOC 2021 codes are precise but not intuitive to search by title alone).

4.3 Aggregation & Trend Modeling

Normalized records are aggregated along four dimensions — company, role family, location, and time period — to produce a trend direction (ramping up / flat / declining) per company-role-location combination. This aggregation layer is the core IP of the platform: the raw data is public, but the normalization and trend logic is not.

4.4 Presentation

v1 ships as a simple search/filter interface: company, role family, location, trend direction. Later phases add alerting, org-structure views, and multi-source confidence scoring per signal.

5. Technical Approach

Layer	v1 Approach	Notes
Ingestion	Python scripts, scheduled quarterly (matches ESDC LMIA class/procedure)	StatCan/SIDAR+ ingestion added in later phase
Storage	PostgreSQL (or SQLite for local prototyping)	Schema: employers, lmia_filings, noc_role_families, aggregate
Normalization	Rule-based + LLM-assisted entity resolution	Start rule-based; add LLM matching for edge cases and N
Backend/API	Lightweight REST API over the aggregate tables	Framework-agnostic; keep it simple for v1
Frontend	Minimal search/filter UI, no auth required for v1	Validate the idea before investing in polish

6. Roadmap

v1 — Core signal engine (current focus)

- Ingest and normalize ESDC Positive LMIA employer data
- Build employer/NOC role family/location/time aggregation
- Ship a basic search/filter UI
- Validate: do real Canadian job seekers find this more useful than a job board?

v2 — Broaden signal coverage

- Add StatCan Job Vacancy and Wage Survey data as a macro validation layer
- Add provincial mass-termination notices where publicly accessible, as a contraction/backfill signal
- Add basic alerting (notify when a tracked employer/role trend changes)

v3 — Public company intelligence

- Integrate SEDAR+ filing language analysis for non-TFWP roles
- Extend coverage beyond TFWP-heavy occupations

v4+ — Engineering signals & org mapping (exploratory)

- GitHub/engineering activity as a leading indicator for tech teams
- Org-structure reconstruction to identify actual hiring-power contacts

7. Risks & Open Questions

- **Field coverage:** v1's core data source (LMIA) only reflects employers who sought to fill roles via the TFWP — strong for tech, trades, and healthcare, weak for many other fields. Does v1 need to explicitly scope its audience, or wait for v2 to broaden?

- **Data lag:** LMIA employer lists are published quarterly, not real-time, so signal freshness has a natural ceiling in v1.
- **Data completeness:** ESDC's published list excludes records containing personal names (e.g. many caregiver-related filings), so coverage is not 100% even within TFWP hiring.
- **Entity resolution accuracy:** Employer name normalization errors directly degrade trend accuracy — this needs ongoing QA, not a one-time fix.
- **Legal/access considerations for provincial layoff data (v2):** no single federal WARN-style system exists in Canada; each province has different rules on what's publicly disclosed, requiring a province-by-province review before scraping at scale.
- **Differentiation risk:** as the idea proves out, will incumbents (LinkedIn, Indeed, Job Bank) replicate the signal? The defensibility is in normalization quality and multi-source fusion, not the raw data itself.

8. Naming & Identity

Name: Prequel — evokes "prequel," i.e. the story that happens before the job posting itself. Chosen to read as a product brand rather than a literal feature description, so it remains fitting even as data sources expand beyond LCA filings.

This is a working reference document, meant to be revised as Prequel moves through each build phase. Update it alongside major scope or architecture decisions.